XRPL Validator — Daily Engineering Report · 2026-08-20

Period: 2026-08-20 00:00 to 24:00 US Eastern (2026-08-20 04:00 to 2026-08-21 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(xrpl-ledger): the flow driver — fib pace, zero-flow TER, consumed-trim stream, dead-level hydration** (`bb1ff88`, xrpl-validator) The whole remaining Payment class (6 both-tesSUCCESS + 3 TER, every multi-strand/pool payment on the board) closed by four receipt-pinned deltas against a skeleton that was already faithful. Design and all receipts in docs/FLOWDRIVER-DESIGN.md.
- **perf(xrpl-node): cache deterministic negative RPC lookups; share the HTTP client** (`e302884`, xrpl-validator) A fully warm single-fixture probe measured 77s wall / 8s CPU — 90% network wait — because the cache stored only SUCCESSFUL lookups, so every negative answer (absent line, absent object: most of what the loaders probe) was a live round-trip to the reference node on every run, and every call built its own reqwest client (no connection reuse).
- **fix(xrpl-ledger): NFT accept — seller-owns-token NoPermission, IOU-priced settlement, strict page bound** (`527b389`, xrpl-validator) The whole NFTokenAcceptOffer cluster, three receipt-pinned rules: 1. "The seller must own the token" (NFTokenAcceptOffer.cpp:229; buy-arm :170): a stale offer whose owner no longer holds the NFToken is tecNO_PERMISSION — our transfer_token surfaced it later as tecNO_ENTRY. #106333939 2FEB03EC accepts a sell offer whose seller parted with 00081B58…0100. Checked in both direct and brokered arms.
- **fix(xrpl-ledger): AMMWithdraw — LP line survives at zero unless default; issuer dir-tail hydration** (`6f72c88`, xrpl-validator)
- A plain withdraw burns LP tokens via redeemIOU — the ORDINARY line machinery — so trustDelete's default-state conditions apply: the LP-side limit must be zero and no quality settings present, else the zeroed LPToken line SURVIVES as Modified. (`deleteAMMTrustLine`, which deletes unconditionally, runs only when the AMM itself is deleted — Transactor.cpp:1050.) Our tear_down_lp_line's own comment stated the …
- **fix(xrpl-ledger): engine-placed offers carry BookDirectory/BookNode/OwnerNode hints** (`2b1fc05`, xrpl-validator) D63363BB (l106030993) + FAFBE1529 (l106065267): a bot places an offer and cancels it six txs later

in the same ledger. Our placement wrote the Offer object WITHOUT its directory hints; OfferCancel reads BookDirectory off the object to unlink the book page, found None, and silently skipped the removal — the emptied page (CE26A7EB...:2) survived. Offers hydrated from mainnet carry the hints, which is why only …

- **fix(xrpl-ledger): four singles — CredentialAccept, SignerListSet, TrustSet, CheckCash** (`7965e2d`, xrpl-validator) CredentialAccept (CredentialAccept.cpp:122-123): expired credential is deleted (both dir unlinks, issuer OwnerCount-1) instead of accepted; on accept the reserve moves issuer->subject when they differ.
- **feat(xrpl-ledger): full PaymentChannel port — tfClose/tfRenew, expiry, keylet, hints** (`56832be`, xrpl-validator) A4E67EFD (l106124864): a pure tfClose claim (Flags only, no Balance) was temMALFORMED because do_apply demanded a Balance. Port the complete PaymentChannelClaim.cpp: expiry-close FIRST (any claimant closes an expired channel, before the permission gate), Balance arm (dst needs a channel authorization — tecNO_PERMISSION under fixCleanup3_2_0 — sig key must match the channel's, tecUNFUNDED_PAYMENT both directions, …
- **feat(xrpl-ledger): MPT subsystem — direct MPT Payments and MPT Clawback** (`121869c`, xrpl-validator) 4F58CD19 (l106134471) and 3EC225FD (l106259185) both died in amount parsing: nothing recognized {mpt_issuance_id, value}. New tx::mpt module ports the MPTokensV1 subset: - keylets: mptIssuance = sha512h(0x007E || MPTID24), mptoken = sha512h(0x0074 || issuance_KEY || holder) — keyed by the issuance's key, not its id. - Payment direct arm (Payment.cpp:516-590; V1 MPT never routes through flow): requireAuth both …
- **feat(xrpl-ledger): anchored-slice bridged legs — single-path pool admission at SPOT, execution capped at the tip** (`4ee4288`, xrpl-validator) #106093637 BDB95F8A, the last parked OfferCreate. tfPassive sell of 1908.4258 RLUSD for 0.02948 BTC, autobridged. After the first pass consumes the 386EAFA9 leg-A head, mainnet's next iteration admits the strand at ub 64717.04 = leg-A tip 1.06832e-6 x leg-B pool SPOT 6.05785e10 (FLOWDBG receipt) and fills 12.03 RLUSD through the rURtT5MM clip offer plus the pool. Our bridged upper bound priced leg B at its FIB rate, …
- **fix(xrpl-ledger): AMM offer generation reads pool balances ZERO-IF-FROZEN** (`29b5ccb`, xrpl-validator) Restock batch 2026-08-20 (24 fresh ledgers): 4 divergent, all ONE family — an arb bot's circular XRP->BITx->REGARD payments (F3BD86FA et al. across l106423982/l106427147/l106428943/l106428898, 24 txs) where mainnet says tecPATH_DRY and we delivered (or tecPATH_PARTIAL'd). The FFI trace names it in rippled's own words: 'AMMLiquidity::getOffer, frozen accounts' — fetchBalances reads through ammHolds(fhZERO_IF_FROZEN), …
- **fix(xrpl-ledger): model ISSUER-ONLY path elements (type 32) as same-currency cross-issuer books** (`603a322`, xrpl-validator) #106425462 CF3FFF81: circular arb XRP → USD.rvYA → USD.rhub8 → XAH.rswh whose path carries {issuer: rhub8, type: 32} between two account hops. rippled's toStrand keeps the running currency and emits a USD.rvYA → USD.rhub8 BOOK step (the trace shows it crossed: offer 576ED819 + two pools) — our path_legs required a currency on every element and dropped the whole path, answering tecPATH_DRY where mainnet delivers 16.328 XAH.
- **fix(xrpl-ledger): payment-mode AMM fills pay the IN-side transfer rate GROSS** (`1e9a2e4`, xrpl-validator) #106426837 6BC6ACF9: circular XRP → FLR → XRPLOL arb through two pools, both tes but 11v8 — every fib fill 0.3% short (ours 1222.247 per slice vs mainnet's 1225.914; the arithmetic pins FLR's TransferRate 1.003 exactly). rippled's rdrIn: the previous step redeems, so the strand parts with ofrIn × rate while the pool receives the net — the trace's 9.931638943875375 = 9.901933144442048 × 1.003 to the digit. Our …
- **feat(xrpl-ledger): last-LP withdrawal deletes the AMM — lines, object, pool account** (`aa37f6b`, xrpl-validator) #106430239 419A5D2C: tfWithdrawAll by the pool's ONLY LP. rippled's deleteAMMAccountIfEmpty

fires once LPTokenBalance hits zero: every pool asset trust line is deleted UNCONDITIONALLY (deleteAMMTrustLine — the survive-at-zero rule is for ordinary withdrawals only), the AMM object leaves the pool's owner directory, and the pool AccountRoot is erased — the trace trustDeletes all three lines then threads to the …

- **fix(xrpl-node): book hydration follows ledger_data markers — 400 pages, budgeted offers** (`64c8246`, xrpl-validator) #106433073 462DE605: a tfImmediateOrCancel selling 4845 drops for RLUSD. The maker's page sat 50 QUALITY LEVELS past the parent tip of the bot-deep RLUSD/XRP book; the trailing enumeration was one 24-row ledger_data call, so the walk saw clob=None and killed an IoC mainnet fills (tecKILLED vs tesSUCCESS, the maker offer well inside our own threshold).

Independence program

No independence record for UTC day 2026-08-20; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Stealth Payment Destinations** (privacy) Add an optional sfStealthScanKey SField to AccountRoot (published via a new AccountSet flag) and a new sfEphemeralPubKey SField on Payment, following
- **Amendment Implementation Bounty Vault** (governance) Add a new ltAMENDMENT_BOUNTY object (owner-reserved, keylet::amendmentBounty(amendmentID)), funded by any account via AmendmentBountyFund into a Singl

- **AMM Concentrated Liquidity Price Bands** (amm-defi) Add a new ltAMM_LIQUIDITY_POSITION object (owner-reserved, keylet::ammPosition(ammID, account, sequence)) and an AMMDepositRange transaction carrying

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-20 05:40	—	xrpl-validator	`bb1ff88`	feat(xrpl-ledger): the flow driver — fib pace, zero-flow TER, consumed-trim stream, dead-level hydration
08-20 15:26	—	xrpl-validator	`e302884`	perf(xrpl-node): cache deterministic negative RPC lookups; share the HTTP client
08-20 15:45	—	xrpl-validator	`527b389`	fix(xrpl-ledger): NFT accept — seller-owns-token NoPermission, IOU-priced settlement, strict page bound
08-20 15:55	—	xrpl-validator	`6f72c88`	fix(xrpl-ledger): AMMWithdraw — LP line survives at zero unless default; issuer dir-tail hydration
08-20 16:56	—	xrpl-validator	`2b1fc05`	fix(xrpl-ledger): engine-placed offers carry BookDirectory/BookNode/OwnerNode hints
08-20 16:56	—	xrpl-validator	`7965e2d`	fix(xrpl-ledger): four singles — CredentialAccept, SignerListSet, TrustSet, CheckCash
08-20 19:08	—	xrpl-validator	`56832be`	feat(xrpl-ledger): full PaymentChannel port — tfClose/tfRenew, expiry, keylet, hints
08-20 19:20	—	xrpl-validator	`121869c`	feat(xrpl-ledger): MPT subsystem — direct MPT Payments and MPT Clawback
08-20 19:45	—	xrpl-validator	`4ee4288`	feat(xrpl-ledger): anchored-slice bridged legs — single-path pool admission at SPOT, execution capped at the tip
08-20 20:41	—	xrpl-validator	`29b5ccb`	fix(xrpl-ledger): AMM offer generation reads pool balances ZERO-IF-FROZEN
08-20 21:26	—	xrpl-validator	`603a322`	fix(xrpl-ledger): model ISSUER-ONLY path elements (type 32) as same-currency cross-issuer books
08-20 21:38	—	xrpl-validator	`1e9a2e4`	fix(xrpl-ledger): payment-mode AMM fills pay the IN-side transfer rate GROSS
08-20 21:43	—	xrpl-validator	`aa37f6b`	feat(xrpl-ledger): last-LP withdrawal deletes the AMM — lines, object, pool account
08-21 03:20	—	xrpl-validator	`64c8246`	fix(xrpl-node): book hydration follows ledger_data markers — 400 pages, budgeted offers

Times in UTC. A commit appears on the day it was committed to main or merged into it.